

Product Requirements Document

Secure Legal Vault

Secure Digital Document Management System for Legal & Investigation Documents

Version: 1.0

Product type: Case-centric, offline-first, tamper-evident document and evidence management platform

Target deployment: Police station/district pilot with scalable integration path for CCTNS, eSakshya, ICJS, e-Courts, e-Prisons, e-Forensics and e-Prosecution.

Confidentiality note: This PRD is a product-design document. It does not replace legal advice, departmental SOPs, official API specifications, or statutory compliance review.

1. Product Overview

1.1 Problem Statement

Law-enforcement and justice agencies manage FIRs, general diary entries, case diaries, witness statements, search and seizure memos, forensic reports, charge sheets, court filings, judgments, digital evidence, and physical evidence records. These materials often remain fragmented across paper files, separate applications, personal devices, and departmental databases.

- Slow document retrieval and inconsistent versions.
- Unauthorized access, credential sharing, and privacy risks.
- Missing or incomplete audit trails.
- Duplicate person records and poor data quality.
- Broken evidence chain of custody.
- Limited rural/offline usability.
- Manual re-entry between police, forensic, prosecution, prison, and court workflows.
- Difficulty proving integrity and provenance of electronic records.

1.2 Product Vision

Build a secure, offline-first digital case vault that preserves document integrity, protects sensitive information, tracks evidence custody, and enables authorized collaboration from FIR registration through investigation, prosecution, court proceedings, and final disposal.

1.3 Core Value Proposition

Stakeholder	Primary value
Investigating Officer	Fast field capture, offline workflows, document retrieval, evidence tracking and charge-sheet preparation.
SHO / Supervisor	Case progress visibility, anomaly alerts, approval workflows and accountability.
Malkhana Officer	QR/Rfid-based evidence intake, movement tracking and inventory audit.
Forensic Personnel	Secure evidence receipt, report upload and return-of-evidence record.
Prosecutor / Court User	Complete, verified and court-ready case material.
Auditor	Tamper-evident access, edit, export, approval and custody records.
Victim / Complainant	Limited, privacy-preserving case milestone view.

2. Goals, Scope and Success Criteria

2.1 Primary Goals

1. Centralize documents and evidence around a single Case ID.
2. Enable secure creation, upload, search, versioning and controlled sharing.
3. Maintain cryptographic integrity for every document version.
4. Support offline field capture and reliable synchronization.
5. Track physical and digital evidence through complete chain of custody.
6. Apply MFA, RBAC, ABAC and dynamic redaction.
7. Detect duplicate person records and link authorized records through a Unique Person ID (UPI).
8. Validate case timelines, authority and procedural rules.
9. Generate court-ready evidence bundles and electronic-record certificate drafts.

10. Integrate with authorized external systems through APIs and webhooks.

2.2 Non-Goals

- The first version does not replace CCTNS, eSakshya, ICJS, e-Courts, e-Prisons, e-Forensics or e-Prosecution.
- The system does not make automated decisions on guilt, innocence, sentencing or criminal risk.
- The system does not expose sensitive case documents to the public.
- The system does not store full sensitive documents directly on a blockchain.
- The system does not force automatic identity merges without human review.
- Any biometric or government identity data requires separate lawful authorization and governance.

2.3 Success Metrics

Category	Metric	Pilot target
Document retrieval	Median time to find requested document	Under 60 seconds
Data quality	Cases with complete mandatory metadata	95%+
Integrity	Uploaded documents with verified lineage	100%
Auditability	Sensitive actions recorded	100%
Evidence	Digitally logged custody movements	95%+
Offline reliability	Offline records synced without loss	99%+
Security	Unresolved critical/high findings before production	0
Adoption	Pilot officers actively using the system	80%+

3. Users and Stakeholders

3.1 User Roles

Role	Key responsibilities / access
System Administrator	Organization, user, device and policy configuration; no implicit unrestricted case access.
Police Station Admin	Local user and device administration within jurisdiction.
Investigating Officer	Create/manage assigned cases, documents, statements, evidence and charge-sheet drafts.
SHO / Supervisor	Assign IOs, review anomalies, approve exceptions and charge sheets.
Data Entry Operator	Create controlled drafts; no approval/finalization rights for restricted documents.
Malkhana Officer	Evidence intake, storage location, transfers, returns, disposal scans.
Forensic Analyst	Evidence receipt/acknowledgement, forensic report upload, return workflow.
Prosecutor	Authorized case review, annotations, missing-item requests and court bundle review.
Court Clerk / Judge	Authorized court views, orders, acknowledgements and verified bundles.
Auditor	Read-only audit/compliance reporting and integrity verification.
Citizen / Victim	Limited milestone status only; no internal case material.

3.2 External Stakeholders

- MHA / NCRB, State Home Department, State Police HQ and IT cell.
- NIC or approved implementation partner.
- District police, prosecution, court registry, prisons, forensic labs and malkhana teams.
- Legal department, cybersecurity teams, privacy/data-governance teams and auditors.

- Victim-support and citizen-service stakeholders.

4. Functional Requirements

4.1 Authentication, Users and Device Security

ID	Requirement	Priority
AUTH-01	MFA login using OTP, authenticator app or certificate-based authentication.	Must
AUTH-02	Role, jurisdiction and case-specific assignment.	Must
AUTH-03	Registered-device requirement for mobile/offline devices.	Must
AUTH-04	Approve, suspend, revoke and remotely disable lost/stolen devices.	Must
AUTH-05	Detect unusual logins, shared-session indicators and repeated failed attempts.	Should
AUTH-06	Optional biometric app unlock on supported managed devices.	Should
AUTH-07	Audit all login, failure, logout, enrolment and revocation events.	Must

4.2 Case Registry and Lifecycle

The platform uses a Case ID as the central link for every person, document, evidence item, forensic report, court action, task and audit event.

ID	Requirement	Priority
CASE-01	Generate a unique Case ID for every case.	Must
CASE-02	Record FIR/GD references, jurisdiction, legal provisions, incident details and assigned officers.	Must
CASE-03	Link persons, documents, evidence, forensic reports, court records and audit events to Case ID.	Must
CASE-04	Provide chronological and category-based case-file views.	Must
CASE-05	Show current stage, pending tasks, due dates, missing documents and anomalies.	Must
CASE-06	Use authorized workflow for reassignment, closure, archival and re-opening.	Must

Case stages: Draft / Preliminary Entry → FIR/GD Registered → Investigation Active → Evidence Collection → Forensic Pending → Charge-Sheet Preparation → Supervisor Review → Filed → Trial → Judgment → Appeal → Closed/Archived.

4.3 Person Registry and Unique Person ID (UPI)

A UPI is an opaque, system-generated identifier for an authorized criminal-justice record. It must not embed personal data and must not be used as a public identity number.

ID	Requirement	Priority
UPI-01	Create UPI after a new person record is validated.	Must
UPI-02	Support person roles: accused, victim, witness, complainant, informant, guardian, expert and others.	Must
UPI-03	Store aliases, language variations, parent/spouse name, age/DOB, address and optional lawful identifiers.	Must
UPI-04	Run duplicate detection before creating a new person.	Must
UPI-05	Show possible matches, confidence score and matching reasons.	Must
UPI-06	Allow existing-UPI link or new-person justification; preserve audit history.	Must
UPI-07	Support human-reviewed merge and split workflow.	Must
UPI-08	Map legacy/local IDs to UPI without deleting original IDs.	Must

Entity resolution inputs: name, aliases, transliteration variants, parent/spouse name, age/DOB, address, phone, case references, vehicle numbers and—for authorized workflows only—lawful biometric/forensic references.

4.4 Document Management

ID	Requirement	Priority
DOC-01	Create, upload, preview, download and organize documents under Case ID.	Must
DOC-02	Support PDFs, images, scanned documents, text, audio and video.	Must
DOC-03	Require document-type-specific metadata.	Must
DOC-04	Create immutable versions; do not overwrite approved/locked content.	Must
DOC-05	Retain original source and generated derivative files separately.	Must
DOC-06	Provide OCR for supported scans and full-text/metadata search.	Should
DOC-07	Support linked persons, evidence, legal sections, tags and confidentiality classification.	Must
DOC-08	Track Draft, Submitted, Reviewed, Approved, Locked, Superseded and Archived states.	Must
DOC-09	Create dynamic redacted views without changing original document.	Must
DOC-10	Log view, print, export, share, edit, approval and delete-attempt actions.	Must

4.5 Document Integrity, Hashing and Digital Signatures

Sensitive source files remain in encrypted storage. The integrity layer stores hashes, version references, actors, timestamps, signatures and event proofs. This creates a tamper-evident record without placing full evidence files on a blockchain.

ID	Requirement	Priority
INT-01	Calculate SHA-256 or approved hash for original upload and each version.	Must
INT-02	Bind hash to document ID, Case ID, version, creator and creation time.	Must
INT-03	Digitally sign critical events: upload, approval, lock, export and evidence transfer.	Must
INT-04	Hash-chain document versions using previous-version hash reference.	Must
INT-05	Store integrity records in append-only, tamper-evident audit storage.	Must
INT-06	Provide authorized Verify Integrity function and clear verification result.	Must
INT-07	Prevent direct content replacement after document lock.	Must
INT-08	Support optional periodic Merkle-root / permissioned-ledger anchoring.	Should

4.6 Evidence Registry and Chain of Custody

ID	Requirement	Priority
EVD-01	Generate unique Evidence ID and link it to Case ID, seizure memo, collector and storage location.	Must
EVD-02	Support physical, digital, documentary, biological, weapon, narcotics, device and media evidence types.	Must
EVD-03	Generate QR code labels; support optional RFID/NFC.	Must
EVD-04	Record custody actions: seize, pack, seal, deposit, transfer, receive, lab dispatch, court production, return, dispose and release.	Must
EVD-05	Record sender, receiver, timestamp, purpose, location and authorization for every transfer.	Must
EVD-06	Require receiver acknowledgement for configured high-risk transfers.	Must
EVD-07	Flag missing steps, overdue returns, invalid movement and unacknowledged handovers.	Must
EVD-08	Provide evidence digital-twin view with current location and full custody timeline.	Must
EVD-09	Support periodic QR/RFID inventory audit.	Should
EVD-10	Support optional smart-locker integration with signed movement authorization.	Could

4.7 OCR, AI Document Intelligence and Knowledge Graph

AI supports officers; it does not make legal determinations. All AI outputs must retain source reference, confidence score and human review controls.

ID	Requirement	Priority
AI-01	Run OCR while preserving original scanned document.	Should
AI-02	Extract candidate entities: people, phones, vehicles, dates, locations, legal sections and evidence IDs.	Should
AI-03	Show confidence and source page/text reference.	Must
AI-04	Allow authorized user to accept, correct or reject extracted data.	Must
AI-05	Suggest related documents/cases based on shared approved entities.	Should
AI-06	Flag possible contradictions or timeline inconsistencies for review.	Should
AI-07	Enforce source-document access policy in graph/search views.	Must
AI-08	Support English, Hindi, Marathi and extensible state-language roadmap.	Should
AI-09	Never automatically alter official document content using AI output.	Must

4.8 Timeline, Authority and Rule Engine

ID	Requirement	Priority
RULE-01	Maintain case timeline for GD, FIR, arrest, search, seizure, remand, forensic and court events.	Must
RULE-02	Validate user authority for charge-sheet submission, approval, transfer and document lock.	Must
RULE-03	Block hard policy violations; allow configured exceptions only with reason and approval.	Must
RULE-04	Flag missing mandatory records before charge-sheet submission.	Must
RULE-05	Show visual timeline with anomalies and reasons.	Should
RULE-06	Calculate case-health score based on completeness, sequencing, authority and custody status.	Should

Initial rules: only assigned IO may submit charge sheet; only authorized supervisor may approve; pre-FIR arrest/seizure requires exception reason and approval; locked charge sheets require amendment workflow; configured evidence transfers require sender/receiver acknowledgement.

4.9 Access Control, Redaction and Audit

ID	Requirement	Priority
ACC-01	Implement RBAC for baseline permissions and ABAC for contextual decisions.	Must
ACC-02	Support agency, jurisdiction, assignment, sensitivity, purpose, time and device-trust attributes.	Must
ACC-03	Support case/document-level grants and time-limited access.	Must
ACC-04	Apply field-level redaction to victim, witness, juvenile and other protected data.	Must
ACC-05	Log access decisions and access denials for high-sensitivity records.	Must
ACC-06	Support emergency break-glass access with reason, enhanced logging and supervisor review.	Should
AUD-01	Log create, read, update, export, print, share, approve, reject, delete-attempt and policy-decision events.	Must
AUD-02	Store audit data: user, role, device, IP/network, time, object, action, result and hash/reference.	Must

4.10 Offline-First Operation and Synchronization

ID	Requirement	Priority
OFF-01	Support offline creation of permitted forms, notes, statements, evidence metadata, photos and documents.	Must
OFF-02	Encrypt local mobile database and protect app with local authentication.	Must
OFF-03	Write every offline action to local append-only audit journal.	Must
OFF-04	Use resumable, queued, incremental/delta synchronization.	Must
OFF-05	Prioritize critical data: FIR, arrest memo, evidence metadata and urgent filing items.	Should
OFF-06	Compress/chunk large media, resume failed uploads and verify final content hash.	Must
OFF-07	Show Offline, Pending, Syncing, Synced, Failed and Conflict states.	Must
OFF-08	Auto-merge only safe low-risk content; route legal-critical conflicts to supervisor.	Must
OFF-09	Do not permit final actions needing online validation unless policy allows deferred workflow.	Must
OFF-10	Optionally support controlled Bluetooth/Wi-Fi Direct peer exchange after security review.	Could

Offline flow: User action → local validation → encrypted local database → local audit hash entry → sync queue → network available? If no, continue offline. If yes, push changes, verify receipt, pull updates and resolve conflicts.

4.11 Electronic Evidence Certificates and Court Bundles

Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 provides for a certificate accompanying electronic records submitted for admission. The product generates a reviewable draft and supports authorized signing; legal officers remain responsible for final legal validation.

ID	Requirement	Priority
CRT-01	Capture source/device metadata, custodian, hash, workflow and custody data where available.	Must
CRT-02	Generate configurable electronic-record certificate draft.	Must
CRT-03	Allow authorized certifying officer review and digital signature.	Must
CRT-04	Generate court bundle: record, certificate, integrity manifest, version history, custody and audit summary.	Must
CRT-05	Attach QR/verification reference and log every bundle export.	Should

4.12 Charge-Sheet Workflow and External Integration

ID	Requirement	Priority
CS-01	Provide charge-sheet workspace linked to documents, persons, evidence and forensic reports.	Must
CS-02	Run completeness, timeline, authority and custody checks before submission.	Must
CS-03	Support IO submit, supervisor approve/reject/return and digitally signed lock.	Must
CS-04	Require amendment/supplementary workflow after lock.	Must
CS-05	Support authorized webhook/API after charge-sheet lock.	Should
CS-06	Transmit minimum necessary approved data and show queued/sent/acknowledged/failed status.	Must

4.13 Citizen Case-Milestone Tracker

ID	Requirement	Priority
CIT-01	Provide verified citizen/victim access to approved high-level milestones only.	Should
CIT-02	Never expose case diaries, witness information, evidence details, internal remarks or restricted orders.	Must
CIT-03	Apply stricter policy or disable view for sensitive categories where required.	Must
CIT-04	Log each citizen lookup.	Must

5. Non-Functional Requirements

5.1 Security

Area	Requirement
Data in transit	TLS 1.3 or approved equivalent.
Data at rest	AES-256 or approved equivalent.
Mobile data	SQLCipher/encrypted SQLite or equivalent.
Password security	Argon2id/bcrypt and enterprise password policy.
Keys	Hardware-backed keystore/KMS/HSM, rotation and revocation.
Signatures	Ed25519/RSA or approved PKI-compatible algorithm.
APIs	OAuth2/OIDC and mTLS for sensitive service-to-service communication.
File storage	Encrypted object storage with strict policy and immutable retention options.
Security operations	Vulnerability scanning, patching, security monitoring and penetration testing.
Least privilege	RBAC + ABAC + case-level access + redaction.

5.2 Performance Targets

Function	Target
Login response	Under 3 seconds on standard network.
Typical search	Under 3 seconds for indexed query.
Case summary load	Under 5 seconds for typical case.
Offline local structured save	Under 1 second.
QR scan custody event	Under 2 seconds under normal condition.
100 MB hash generation	Under 5 seconds on supported server/device profile.
Media upload	Resumable with visible progress and final hash verification.
Sync recovery	Resume from last confirmed event/chunk.

5.3 Reliability, Usability and Accessibility

- Pilot central-service availability target: 99.5%, excluding scheduled maintenance.
- Offline-permitted activities must continue during connectivity outage.
- No locally saved action may disappear without visible error and audit event.
- Backups must be encrypted, tested and recoverable.
- Android-first field UX with readable typography, clear status, low cognitive load and simple language.
- Initial language support: English, Hindi and Marathi; extensible for additional Indian languages.

- Officer-facing UI must avoid unnecessary technical terms such as blockchain, Merkle root or cryptographic ledger.

6. High-Level Architecture

User Interfaces (Web Portal, Mobile App, Citizen Milestone Portal) → Identity and Access Layer (MFA, RBAC, ABAC, PKI) → Core DMS Platform (Case Registry, Person/UIP, Documents, Evidence, Search, Redaction, Court Bundles) → Integrity & Intelligence Layer (Hashing, Signatures, Audit Ledger, OCR/NLP, Entity Resolution, Rule Engine) → Storage & Offline Layer (PostgreSQL, Encrypted Object Storage, Mobile SQLite/SQLCipher, Sync Queue) → Integration Layer (API Gateway, Adapters, Webhooks).

6.1 Minimal Technology Stack

Layer	Recommended stack
Web frontend	React.js + TypeScript + Material UI
Mobile	Flutter or React Native; Android-first
Backend	FastAPI (Python) or NestJS (Node.js)
Primary database	PostgreSQL
File storage	S3-compatible encrypted object storage / approved government cloud equivalent
Offline storage	SQLite + SQLCipher
Search	PostgreSQL full-text initially; OpenSearch/Elasticsearch at scale
Queue	RabbitMQ or Kafka
OCR / NLP	Tesseract/EasyOCR + Indic-language capable NER; human review mandatory
Cryptography	SHA-256 + Ed25519/RSA + PKI/KMS integration
Deployment	Docker; Kubernetes for scale
Monitoring	Prometheus + Grafana; OpenSearch/ELK-style logging

7. Core Data Model

7.1 Primary Entities

User, Role, Organization, Jurisdiction, Device, Case, CaseAssignment, Person, PersonAlias, UIP_Mapping, Document, DocumentVersion, DocumentMetadata, EvidenceItem, EvidenceCustodyEvent, ForensicReport, TimelineEvent, WorkflowTask, Approval, AccessPolicy, AccessGrant, RedactionRule, AuditEvent, IntegrityEvent, SyncQueueItem, OfflineJournalEvent, Certificate, CourtBundle, IntegrationEvent, Notification, RetentionPolicy and LegalHold.

7.2 Primary Relationships

Case 1:N Document; Case 1:N EvidenceItem; Case N:N Person through CasePersonRole; Case 1:N TimelineEvent; Document 1:N DocumentVersion; DocumentVersion 1:N IntegrityEvent; EvidenceItem 1:N EvidenceCustodyEvent; User 1:N AuditEvent; User N:N Case through CaseAssignment.

8. Key User Stories

- As an IO, I want to create a case and upload FIR-related records so all material is in one location.
- As an IO, I want to capture a witness statement offline so field work continues during network failure.
- As an IO, I want duplicate-person suggestions before new UIP creation so I do not create duplicate identities.
- As a malkhana officer, I want to scan QR labels at intake and movement so I maintain custody accurately.
- As a supervisor, I want to see missing documents, timeline anomalies and overdue evidence returns so I can intervene early.

- As a prosecutor, I want a verified, searchable case file so I can prepare filings efficiently.
- As an auditor, I want to see who accessed, changed, printed or exported sensitive content.
- As a verified victim/complainant, I want basic case milestones without exposure of confidential investigation data.

9. MVP Scope

9.1 Must-Have MVP Modules

1. MFA simulation and roles: IO, SHO, Malkhana Officer, Prosecutor, Judge/Clerk, Admin.
2. Case creation and FIR/document upload.
3. Person creation, duplicate check and UPI generation.
4. Document metadata, version history, SHA-256 hashing and Verify Integrity screen.
5. Tamper-evident audit log.
6. Offline witness statement capture with local encrypted store and sync queue.
7. Evidence registration with QR code and custody event log.
8. Timeline rule checks: arrest/FIR ordering, IO-only charge-sheet submission, supervisor approval.
9. Charge-sheet review, approval and lock simulation.
10. Court-bundle PDF export with certificate draft and audit summary.
11. Dashboard for active cases, anomalies, pending sync and audit events.

9.2 MVP Demo Scenario

1. IO logs in with MFA.
2. IO creates an FIR/case and links or creates UPI for accused/victim.
3. IO captures witness statement offline; it is saved locally as Sync Pending.
4. Connectivity returns; statement and audit events synchronize.
5. IO registers seized mobile phone, system generates Evidence ID and QR label.
6. Malkhana officer scans QR and acknowledges custody.
7. Forensic analyst uploads report.
8. IO prepares charge sheet; rule engine flags a missing approval/timeline issue.
9. IO fixes issue; supervisor approves and system cryptographically locks final version.
10. System exports court bundle with integrity proof, custody summary and certificate draft.
11. System displays integration event as Ready for authorized ICJS/e-Courts API transmission.

10. Implementation Roadmap

Phase	Duration	Main deliverables
Phase 0: Discovery & Governance	2–4 weeks	Stakeholder map, workflow mapping, taxonomy, privacy/threat model, integration inventory, baseline KPIs.
Phase 1: Core Secure DMS	6–8 weeks	MFA, user/device management, case registry, documents, search, versioning, hashing, audit, basic RBAC.
Phase 2: Offline & Evidence	6–8 weeks	Encrypted local DB, sync queue, local journal, evidence QR workflow, custody events, ABAC/redaction, approvals.
Phase 3: Intelligence & Court Readiness	6–8 weeks	UPI/entity resolution, OCR, rule engine, case-health score, certificate drafts, court bundles, graph POC.
Phase 4: Integration & Pilot	8–12 weeks	API gateway, mock/approved adapters, monitoring, pilot migration, training and security hardening.

11. Testing Strategy

11.1 Functional Testing

- Case, person, document, evidence, QR scan, workflow, court bundle, UPI, timeline rule and redaction tests.
- Offline capture, upload resume, sync retry, conflict resolution and local-journal validation.
- Search and metadata validation across supported document types.

11.2 Security Testing

- Authentication bypass, privilege escalation, unauthorized document access, API authorization, session/token misuse and device revocation.
- File-upload validation, malware scanning, SQL injection/XSS checks and logging verification.
- Hash/signature verification and audit-log tamper simulation.
- Independent penetration test before production deployment.

11.3 Field and Performance Testing

- Airplane mode, unstable 2G-like network, interrupted upload, device restart, low storage and concurrent-edit scenarios.
- Large-file upload, high-volume audit events, search performance, sync queue behaviour and OCR queue load testing.

12. Risks and Mitigation

Risk	Mitigation
Production API access unavailable	Use mock/sandbox adapters; define API contracts; preserve connector architecture.
Legal uncertainty for electronic evidence	Validate certificate templates and processes with authorized legal/prosecution stakeholders.
False UPI match	Human review, confidence thresholds, merge/split workflow and immutable decision log.
Shared credentials	MFA, device binding, session anomaly detection and training.
Offline device loss	Encryption, app lock, minimized cache, remote revocation/wipe and key rotation.
OCR errors	Preserve originals; show confidence/source; require human review.
Blockchain complexity	Use hash-chained/WORM ledger in MVP; introduce permissioned ledger only if justified.
Large media files	Compression, chunking, resumable uploads, priority sync and object storage.
Low field adoption	Simple UX, offline reliability, co-design, training and rapid support.
Uncontrolled sharing	ABAC, redaction, data minimization, audit alerts and formal sharing policy.

13. Compliance and Governance

- The platform must align with authorized departmental SOPs, applicable criminal procedure, evidence law, privacy obligations and records-retention rules.
- Electronic-record certificates must be reviewed and approved by competent legal authority before operational use.
- The system must preserve originals, timestamps, signatures, custody events and audit records.
- Victims, juveniles, protected witnesses, sexual-offence cases and national-security cases must receive enhanced access controls.
- Data sharing must follow minimum-necessary-data principles, formal authorization and approved APIs.
- AI recommendations must be explainable, reviewable and never the sole basis for a legal or disciplinary decision.
- Biometric or government identity integration requires separate legal authorization, security controls and governance.

14. Future Enhancements

- RFID/NFC evidence tags and smart evidence lockers.
- Secure inter-agency virtual data rooms.
- Multilingual speech-to-text and advanced handwriting OCR.
- Knowledge-graph based related-case discovery.
- Geofenced pre-caching and controlled P2P field mesh sync.
- DigiLocker integration for authorized certificate delivery.
- Public certificate-authenticity verification portal.
- Advanced forgery detection for scanned documents.
- Automated retention/disposal and state/district compliance analytics.

15. Final Acceptance Criteria

1. Authorized users can manage cases and documents through a case-centric workspace.
2. Every upload/version has verifiable hash lineage and audit record.

3. Field users can save permitted records offline and synchronize later without data loss.
4. Sensitive content is redacted for unauthorized users.
5. Evidence can be registered, QR-tagged, transferred and audited end-to-end.
6. Unauthorized charge-sheet submission is blocked and defined anomalies are flagged.
7. A court bundle includes evidence/document, integrity manifest, custody summary, audit summary and certificate draft.
8. UPI creation and duplicate review work with human confirmation.
9. Typical document retrieval is under one minute in pilot conditions.
10. Security, usability, legal and stakeholder review are completed before scale-up.

16. Reference Context

This PRD is designed to complement India's digital criminal-justice ecosystem. ICJS is intended to integrate police/CCTNS, courts/e-Courts, prisons/e-Prisons, forensics/e-Forensics and prosecution/e-Prosecution. eSakshya supports digital evidence collection and preservation workflows. Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 is relevant to certification of electronic records. Official specifications, policy and legal review must be obtained before deployment.

- MHA: Inter-Operable Criminal Justice System (ICJS) framework.
- MHA / NCRB: CCTNS and criminal-justice integration context.
- NIC: eSakshya evidence collection and preservation context.
- Bharatiya Sakshya Adhiniyam, 2023: electronic-record certification context.